

MID TERM EXAMINATION

Paper: Mid Term Examination - Set A | Duration: 3 Hours | Max Marks: 84 | Date: 14-03-2026

Total Questions: 18 | Answer all questions.

Name: _____

Roll No: _____

Student Sign: _____

Teacher Sign: _____

SECTION A: MCQS

1. Which of the following best describes an Abstract Data Type (ADT)? [2 Marks]
 - a) A data structure that only stores abstract values.
 - b) A set of data values and associated operations, without specifying implementation details.
 - c) A concrete implementation of a data structure.
 - d) A data type that can only be accessed through pointers.
2. What is the worst-case time complexity for inserting an element at the beginning of a singly linked list? [2 Marks]
 - a) $O(1)$
 - b) $O(\log n)$
 - c) $O(n)$
 - d) $O(n \log n)$
3. A data structure where elements are added at one end and removed from the other end is known as a: [2 Marks]
 - a) Stack
 - b) Queue
 - c) Tree
 - d) Graph
4. Which of the following is a disadvantage of using arrays compared to linked lists? [2 Marks]
 - a) Faster access to elements by index.
 - b) Requires contiguous memory allocation.
 - c) More complex to implement.
 - d) Less memory efficient for a fixed size.
5. If a stack has a capacity of 5 elements and it currently contains 3 elements, what would be the result of a 'push' operation followed by a 'pop' operation? [2 Marks]
 - a) The stack size remains 3, and the pushed element is returned.
 - b) The stack size becomes 4, and the last pushed element is returned.
 - c) The stack size remains 3, and the element pushed before the last one is returned.
 - d) The stack size becomes 2, and an error occurs.
6. Briefly explain the purpose of Big O notation in algorithm analysis. [3 Marks]

SECTION A: SHORT ANSWERS

7. Define a Data Structure. [3 Marks]
8. State one primary advantage of a linked list over an array for dynamic data storage. [3 Marks]
9. Name two common applications of queues. [3 Marks]

SECTION A: LONG ANSWERS

10. What does the acronym "LIFO" stand for in the context of data structures? [3 Marks]

SECTION A: MCQS

11. Describe the steps involved in inserting a new node at the beginning of a singly linked list. [5 Marks]

SECTION A: SHORT ANSWERS

12. Differentiate between linear and non-linear data structures, providing an example for each. [5 Marks]
13. Explain how a Circular Queue addresses the limitation of "unutilized space" often encountered in a Linear Queue implementation using an array. [5 Marks]
14. List and briefly explain the primary operations on a Stack Abstract Data Type (ADT). [5 Marks]

SECTION A: LONG ANSWERS

15. Explain the concepts of Time Complexity and Space Complexity in the context of algorithm efficiency. [5 Marks]

SECTION A: MCQS

16. Design and explain the steps for implementing a basic Stack ADT using an array. Include the algorithms for push, pop, and peek operations. Clearly state any assumptions. [10 Marks]
17. Compare and contrast Arrays and Singly Linked Lists based on memory allocation, access method, insertion/deletion efficiency, and storage overhead. Provide scenarios where one is preferred over the other. [12 Marks]

SECTION A: SHORT ANSWERS

18. Explain the concept of a Doubly Linked List. Describe the algorithm for deleting a specific node (given its data value) from a Doubly Linked List. Illustrate with a conceptual diagram. [12 Marks]
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